

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

QUESTIONS: 5 Total Marks: 50 Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I **Samuel V(192412237)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Samuel V

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication

fails. Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
 - Optimizing the IPv6 configuration by assigning appropriate addresses.
 - Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication

fails. Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
 - Optimizing the IPv6 configuration by assigning appropriate addresses.
 - Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination	Network	Next Hop
192.168.1.0/24	10.0.0.2	
192.168.2.0/24	10.0.0.3	Default Route 10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network. Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

Answer:

Q3. Subnet Allocation Debugging and VLSM Optimization Using Cisco Packet Tracer

Figure 1: Initial Network Topology

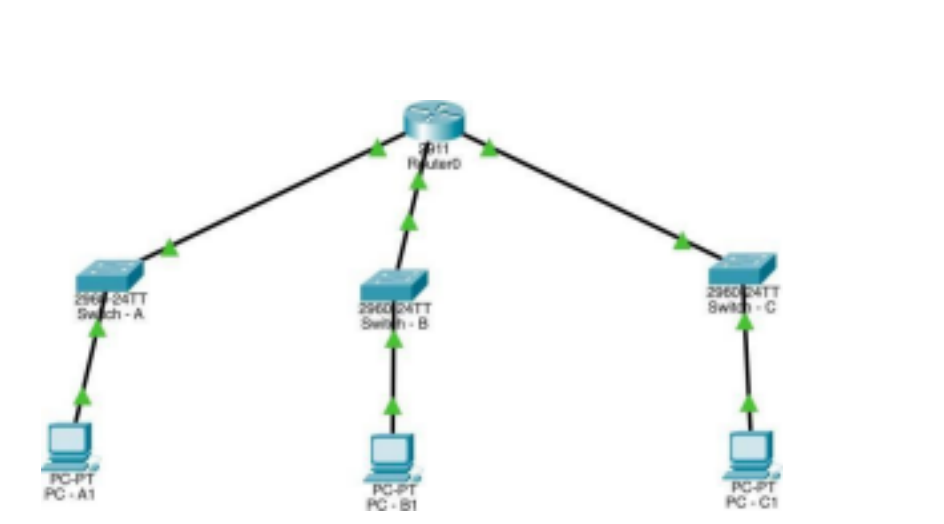


Figure 2: Initial router interface configuration for the existing subnet allocation.

```
Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       192.168.1.1     YES manual up          up
GigabitEthernet0/1       192.168.1.65    YES manual up          up
GigabitEthernet0/2       192.168.1.97    YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
Router#
```

Figure 3: IP configuration of PC-A1 using the /26 subnet.



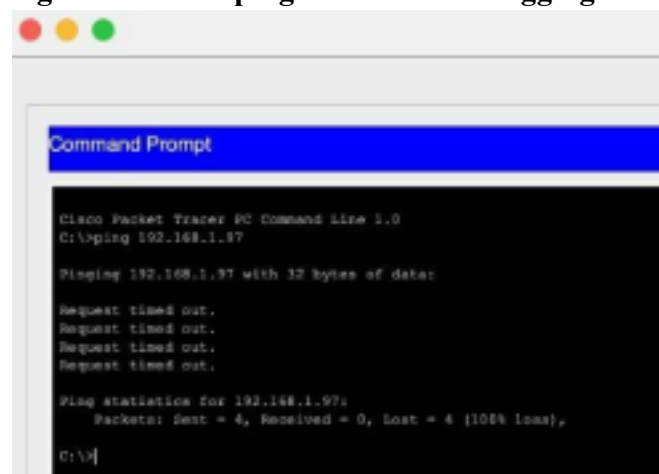
Figure 4: IP configuration of PC-B1 using the /27 subnet.



Figure 5: Initial IP configuration of PC-C1 using the /28 subnet.



Figure 6: Failed ping test before debugging due to an addressing conflict.



Debugging Finding: The same IP address, 192.168.1.70, was assigned to two different PCs, causing an IP address conflict.

Figure 7: Corrected IP configuration of PC-C1 after resolving the duplicate IP address.



Figure 8: Successful ping test after correcting the IP addressing error.

```
C:\>ping 192.168.1.97

Pinging 192.168.1.97 with 32 bytes of data:

Reply from 192.168.1.97: bytes=32 time<1ms TTL=255
Reply from 192.168.1.97: bytes=32 time<1ms TTL=255
Reply from 192.168.1.97: bytes=32 time<1ms TTL=255
Reply from 192.168.1.97: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.1.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

RUBRICS FOR CALCULATION

			Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors
Criteria	Weightage	Level (Exem y)				
Problem Identification	20%	Accurately identifies all errors and root causes clearly understanding				
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained	
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated	

Code Quality	15%	Produces clean, well structured code with proper comments and documenta tio n	Code is mostly clear with minor documen tat ion gaps	Code lacks structure and proper documenta tio n	Poorly written code with no document ati o n
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